

Urine-powered socks can send a text

Robotics professor Ioannis Ieropoulos and his team at University of the West of England in Bristol, UK, have managed to turn human waste and locomotion into electrical power with a bit of help from microbes.



Urine-powered socks (Image courtesy: www.twitter.com)

According to the scientists, this is the first time anyone has combined microbial fuel cells with wearable technology. The socks produced enough electricity to power a specialised wireless transmitter that sent out the message 'World's First Wearable MFC' every two minutes.

The idea behind the creation was to create a self-contained system for generating power, with an eye toward survivalist scenarios.

Self-healing gel for foldable electronics

Researchers at University of Texas, Austin, USA, have developed a self-healing gel that repairs and connects electronic circuits, paving the way for the development of flexible electronics.

The new material has high conductivity, and strong mechanical and electrical self-healing properties.

Until now, self-healing materials have relied on application of external stimuli such as light or heat to activate repair.

The research team created the self-healing gel by combining two gels: a self-assembling metal-ligand gel that provides self-healing properties and a polymer hydrogel that is a conductor.

To construct the self-healing electronic circuit, the gel would not replace the typical metal conductors that transport electricity, but could be used as a soft joint, joining other parts of the circuit.

3D-printed rotors that freeze seawater into drinking water

GE has long been an expert in steam turbine technology for desalination of water, but in 2015, it began to miniaturise the process.

A press release from the company explains, "As part of

the water desalination technology being developed with the US Department of Energy, researchers are using the same steam turbine turbomachinery 3D printed in a miniaturised form to compress and stream a mixture of air, salt and water through a hyper-cooling loop that freezes seawater. By freezing the mixture, the salt naturally separates in solid form, leaving just ice. The ice is then melted, leaving clean water."

The design is a low-cost, low-energy way of creating drinking water, and GE will continue to test the technology through mid-2016 to assess its feasibility.

Renesas GR Kaede Design Contest conducts grand finale in Bengaluru

Renesas GR Kaede Design Contest 2015, the second edition of the popular design contest organised by Renesas and EBY, conducted the grand finale at Shangri-La Hotel in Bengaluru on December 16, 2015. The top-three projects



Renesas contest winners

were on sign-language recognition using sensor gloves, smart agriculture system and Upasana diagnostic toolkit for Asha workers. The most innovative project was awarded to ATM Transactions Using Iris-Recognition Technology. Final winners will be announced at DevCon 2016, Bengaluru, to be held in January.

Thermally-durable lithium-ion battery reduces thermal resistance

Hitachi Ltd and Tohoku University's Advanced Institute for Material Research group have developed a basic technology to reduce the internal resistance of all-solid-state lithium-ion batteries using a complex hydride as a solid electrolyte under the project titled 'Collaborative Research for Next Generation Innovative Battery.'

With reduced internal resistance, charge-discharge performance of the battery improves and can well operate at temperatures as high as 150°C. The battery does not require a cooling system, which results in reduction of overall costs.

The new battery was found to have a discharge capacity of 90 per cent of theoretical value from a prototype of the lithium-ion battery with the capacity of 2mAh and the energy density of 30Wh/L during experiments.